DS lab 3.b .c - Code::Blocks 25.03

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```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* top = NULL;
void push(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    if (newNode == NULL) {
        printf("Stack Overflow (Memory not available)\n");
        return;
    }
    newNode->data = value;
    newNode->next = top;
    top = newNode;

    printf("%d pushed to stack\n", value);
}
void pop() {
    if (top == NULL) {
        printf("Stack Underflow (Empty stack)\n");
        return;
    }
    struct Node* temp = top;
    printf("%d popped from stack\n", top->data);

    top = top->next;
    free(temp);
}


void peek() {
    if (top == NULL) {
```

C:\Users\Sevanthi-v\Downloads\DS lab 3.b .c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 1, Col 1, Pos 0 | Insert | Read/Write | default

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```c
void peek() {
    if (top == NULL) {
        printf("Stack is empty\n");
    } else {
        printf("Top element is: %d\n", top->data);
    }
}


void display() {
    if (top == NULL) {
        printf("Stack is empty\n");
        return;
    }

    struct Node* temp = top;
    printf("Stack elements: ");

    while (temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}


int main() {
    int choice, value;

    while (1) {
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n2. Pop\n3. Peek\n4. Display\n5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

```

C:\Users\Sevanthi-v\Downloads\DS lab 3.b .c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 32, Col 1, Pos 695 | Insert | Read/Write | default

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```c
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n2. Pop\n3. Peek\n4. Display\n5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter value: ");
                scanf("%d", &value);
                push(value);
                break;

            case 2:
                pop();
                break;

            case 3:
                peek();
                break;
            case 4:
                display();
                break;
            case 5:
                exit(0);
            default:
                printf("Invalid choice\n");
        }
    }
    return 0;
}
```

C:\Users\Sevanthi-v\Downloads\DS lab 3.b .c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 32, Col 1, Pos 695 | Insert | Read/Write | default

```
--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value: 13
13 pushed to stack

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 2
13 popped from stack

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 16
Invalid choice

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
```

```
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Invalid choice

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 3
Stack is empty

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack is empty

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 5

Process returned 0 (0x0)   execution time : 24.204 s
Press any key to continue.
```